A Condition-Indexed Operating Envelope and Representation-Repair Study for Automatic Modulation Classification under Structured Interference

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***Abstract*—Uniform model rankings hide a central design question in automatic modulation classification (AMC): under structured interference, the ranking of a compact spectral representation and a much larger I/Q-domain classifier can itself depend on physical conditions. We study this rank exchange in source-disjoint 3GPP TR 38.901 TDL-profile simulations, comparing VIMD-Net, a 39500-parameter spectral network, with strong I/Q-domain comparators across the complete SNR–SIR plane. The original sealed confirmatory family resolves a positive whole-configuration contrast of 4.57 percentage points ($[3.65, 5.49]$ percentage points) for the complete configuration over its shared spectral backbone, while the two component-level interventions (teacher form and route count) are bounded rather than individually resolved. Marginally over the hard-interference split the compact model trails both I/Q comparators (macro-F1 0.4406 versus 0.4599 and 0.4872); the ranking reverses only in the severe, high-overlap corner. In a post-hoc severe-interference analysis at SIR $= -15$ dB, the compact model gains 9.74 percentage points over MCLDNN and 10.46 percentage points over IQFormer-inspired, with positive paired intervals and the same direction in every seed. A two-variable overlap–SIR envelope, where overlap denotes the fraction of jammer energy falling inside the target's dominant time–frequency support, is fitted on in-distribution and hard-interference cells only. It predicts the gain against IQFormer-inspired on held-out regime cells with RMSE 6.20 percentage points versus 8.40 percentage points for a constant predictor estimated from the fitting cells ($R^2_{\text{skill}} = 0.46$); the validated skill is cross-regime transport of gain magnitude rather than sign classification. A failed clean-retention gate is then localized and repaired prospectively: a received-I/Q sidecar raises the total model size from 39500 to 46794 parameters while improving clean macro-F1 by 11.77 percentage points and hard-interference macro-F1 by 4.37 percentage points; these Tier-2 results are exploratory and outside the sealed family. The paper reports a simulation-only operating envelope and a repair path, not uniform superiority.**

***Index Terms*—Automatic modulation classification, structured interference, interference overlap, vehicular Doppler, operating envelope, condition-dependent model comparison, representation repair, compact neural networks.**

Evidence status. The original sealed family resolves one positive whole-configuration contrast (A5 vs A0), while two component contrasts and the clean-retention gate do not pass. Severe-interference, operating-envelope, and Tier-2 repair results are exploratory or post-hoc. All conclusions are restricted to source-disjoint simulation.

I. Introduction

Automatic modulation classification supports spectrum monitoring and blind receiver configuration by inferring a waveform's modulation from received samples. Deep AMC has progressed from convolutional I/Q classifiers [1], [2] to complex-valued, multi-stream, recurrent, and Transformer-like representations [3]–[7]. Compact architectures remain attractive when inference cost matters [8], [9], but average accuracy alone does not say where a compact representation is preferable.

This omission is consequential in mobile and vehicular receivers. A high-mobility terminal facing structured interference (a sweep, tone, pulse, partial-band process, or modulated out-of-taxonomy emitter overlapping the target in time and frequency) must decide on a limited compute budget whether to run a compact spectral front end or a high-capacity I/Q path. The same model can trail a high-capacity I/Q classifier in weak or narrowband interference yet lead it in a severe corner where the interference overlaps the target's own time–frequency support. Averaging these regimes produces a ranking but erases the design rule.

This paper asks a narrower question: *can the sign and size of the compact-model advantage be indexed by physical condition variables, and can the boundary revealed by that index be repaired?* We answer using a frozen source-disjoint simulation campaign, artifact-only post-hoc analysis, and prospective Tier-2 experiments. The base model, VIMD-Net, maps a complex STFT to target-dominant, jammer-dominant, and unexplained-or-ambiguous routes; its supervision is physics-informed in the sense that a simulation-component teacher, available only during training, is derived from independently tracked target, jammer, and residual components. We do not claim that a specific routing or teacher component is individually confirmed (Section V-A); the contribution is the boundary and its repair, not the architecture itself.

The contributions are:

- a condition-indexed rank exchange, exposed over the complete structured-interference SNR–SIR plane, between a compact spectral model and stronger I/Q-domain classifiers, instead of a single marginal ranking;
- a low-dimensional target-support-overlap–SIR empirical envelope whose cross-regime gain-transport skill is evaluated on regime cells excluded from fitting, against an explicit fit-domain constant baseline and with a bounded, non-causal interpretation;
- a separation of the original sealed whole-configuration contrast from post-hoc severe-interference evidence, using paired statistics, multiplicity control, and full-region visualization so a favorable corner is not read as uniform superiority; and
- a prospective diagnosis-and-repair chain that turns a failed clean-retention gate into a representation-information finding: a received-I/Q sidecar increases the total model size only modestly while repairing clean and

hard-interference performance, and a spectral capacity ladder shows that width alone does not remove the boundary.

The claim is deliberately not that VIMD-Net is uniformly best. It is that the compact and high-capacity representations exchange rank in a reproducible, condition-dependent way, that this boundary can be described by physical covariates, and that, for its principal clean boundary, the evidence is more consistent with a repairable representation-information gap than with insufficient clean exposure alone.

II. Related Work and Claim Boundary

A. Strong and Efficient AMC Representations

CNN and recurrent AMC systems learn directly from I/Q sequences [1], [2], [5]; later work preserves complex structure, fuses temporal and spatial streams, or applies attention [3], [4], [6], [7]. Vehicular and mobile receivers have motivated a distinct line of work on reparameterized causal convolutions, cross-scale kernel selection with IQ-imbalance and carrier-offset compensation, and channel-robust cumulant features [10]–[12]. Compact and efficient AMC has received sustained attention for cost-sensitive deployment [8], [9]. These lines of work motivate strong I/Q-domain comparators and prevent us from presenting a lightweight architecture or a fusion scheme as the primary novelty.

B. Robust and Cross-Domain AMC

Channel adaptation, masked or contrastive learning, and interference-tolerant features address different nuisance sources [13]–[16]. Overlapped-signal classification, denoising masked autoencoding, and cross-domain prototypical adaptation further populate the 2024–2026 frontier [17]–[19]. These works establish that “robust AMC” and “cross-domain AMC” are mature themes; we therefore do not claim robustness or adaptation as novelty.

C. Interference-Aware and Malicious-Interference AMC

Interference-tolerant recognition [16] and tensor-based malicious-interference recognition for MIMO [20] already address structured interference directly, including low-SIR conditions. Our setting differs: we do not perform source reconstruction or tensor separation, and we do not aim to be a universally interference-robust classifier. We study how the *ranking* of compact spectral versus high-capacity I/Q models changes with the interference condition itself.

D. Distinction: Condition-Indexed Ranking and Repair

Existing work has made substantial progress in interference-tolerant recognition, domain adaptation, lightweight architectures, and multi-representation fusion (including recent knowledge-guided and multimodal preprints [21]–[24]). Our contribution is not another claim that a particular representation dominates on average. We study the condition dependence of the ranking itself, fit a low-dimensional empirical boundary to that rank exchange, expose the negative regions rather than reporting only favorable averages, and prospectively intervene on the revealed representation boundary. To our knowledge, prior AMC studies have largely optimized or compared marginal performance rather than explicitly modeling the condition-dependent rank exchange between compact spectral and high-capacity I/Q representations and then intervening on the boundary; we state this as a positioning claim, not an exhaustive systematic-review result.

TABLE I
Coherence-Time Bookkeeping ($T_c = 0.423/f_D$ at $f_c = 5.9$ GHz).

Speed (km/h)	f_D (Hz)	T_c (μs)	T_c / window
60	328	1290	1.26
120	656	645	0.63
150 (training max)	820	516	0.50
180 (held)	984	430	0.42
250 (held max)	1367	309	0.30

III. System Model and VIMD Configuration

A. Mobile Received Signal and Paired Views

We model a receiver-side AMC front end under terrestrial high-mobility conditions. For source sequence $s[n]$ and jammer $u[n]$, one received window is

$$x[n] = \mathcal{R}(\mathcal{H}_s(s)[n] + \kappa\mathcal{H}_u(u)[n] + w[n]) , \tag{1}$$

where $\mathcal{H}_s$ and $\mathcal{H}_u$ are independently drawn fading and mobility operators (parameterized by carrier frequency, terminal speed, and 3GPP TR 38.901 [25] TDL-A/C/D profiles during training, with TDL-B/E held out for channel evaluation), κ scales the jammer contribution so that the realized SIR, defined as $10\log_{10}\big(E[|\mathcal{H}_s(s)|^2]/E[|\kappa\mathcal{H}_u(u)|^2]\big)$ on the simulator’s tracked powers, matches the target value, w is receiver noise, and $\mathcal{R}$ applies the frozen receiver chain. The carrier frequency is $f_c = 5.9$ GHz, representative of intelligent-transport-system (ITS) and vehicle-to-everything (V2X) operation [26], and the complex baseband sampling rate is 1 MHz. Terminal speeds seen during training are 0, 60, 120, and 150 km/h, giving a maximum nominal Doppler shift of about 820 Hz; the held-speed split uses 180 and 250 km/h, i.e. up to about 1367 Hz, so speed and TDL profile are held out along separate axes. These speeds justify the high-mobility label quantitatively rather than rhetorically: using the conventional approximation $T_c \approx 0.423/f_D$, Table I lists the coherence time at each training and held-out speed. Within the training range the channel is approximately static at 60 km/h (1290 μs) but drops below the 1024 μs decision window from 120 km/h upward (645 μs), reaching 516 μs at the 150 km/h training maximum; the held-out speeds extend this to 430 and 309 μs at 180 and 250 km/h. The held-speed split therefore probes deeper intra-window non-stationarity in addition to a Doppler-spread shift, so unseen speed is a genuine regime change rather than a rhetorical label.

Training pairs two independently impaired observations of the same immutable source; inference uses one view. Pairing is therefore a training and statistical-alignment device, not additional test-time information. This model is a controlled simulation of a mobile receiver, not a complete V2X geometry; no MAC- or network-layer V2X performance is claimed.

Unanchored in-taxonomy cochannel collisions are excluded: without a physical target anchor, exchanging two admissible emitters can preserve the mixture while changing the requested single label, which requires a different label definition.

B. Three-Route Spectral Allocation

Let $Z = \mathrm{STFT}(x) \in \mathbb{C}^{F\times T}$. The front end receives real, imaginary, and log-magnitude planes and produces logits A_s, A_j, A_o on the same lattice. The allocation masks are

$$\begin{aligned} &[M_s, M_j, M_o] = \mathrm{softmax}([A_s, A_j, A_o]),\\ &M_s + M_j + M_o = 1. \end{aligned} \tag{2}$$

An environment encoder produces a scalar ambiguous-route gate λ and a bounded bypass ρ. The modulation and jammer routes are

$$Z_m = (M_s + \lambda M_o + \rho) \odot Z, \tag{3}$$

$$Z_j = (M_j + (1-\lambda) M_o) \odot Z. \tag{4}$$

Because these weights are real and nonnegative, every retained coefficient keeps the mixture STFT phase. The bounded bypass ρ is a per-window scalar in the frozen range $[0.05, 0.35]$; because it is added to both route weights, the allocation stays nonnegative but is no longer an exact partition of the mixture power, i.e. $Z_m + Z_j \neq Z$ in general. This is the ratio-masking construction familiar from supervised source separation [27], [28], where the limits of magnitude-only masks and the value of phase-aware or complex targets are well documented [29], [30]; the elementary phase-preservation property does not imply source identifiability.

C. Simulation-Component Teacher

During training only, independently tracked post-channel target, jammer, and residual components define powers P_s, P_j, P_u on the student's lattice. With $q_k = P_k/(P_s + P_j + P_u)$, the fixed teacher is

$$M_s^\star = [q_s - q_j]_+, \qquad M_j^\star = [q_j - q_s]_+, \tag{5}$$

$$M_o^\star = q_u + 2\min(q_s, q_j). \tag{6}$$

The routes are nonnegative and sum to one; negligible-energy cells are assigned to the uncertainty route by the finite-precision implementation. We call this construction *physics-informed* because the supervision uses simulator-tracked components; it does not imply that the margin teacher or the three-route allocation is individually confirmed (Section V-A). The teacher is not needed at inference.

D. Training Objective

The total loss is

$$\begin{aligned} \mathcal{L} = \mathcal{L}_{\mathrm{mod}} &+ \alpha\mathcal{L}_{\mathrm{teacher}} + \beta\mathcal{L}_{\mathrm{jammer}} + \gamma\mathcal{L}_{\mathrm{link}}\\ &+ \eta\mathcal{L}_{\mathrm{contrast}} + \theta\mathcal{L}_{\mathrm{orth}}, \end{aligned} \tag{7}$$

where $\mathcal{L}_{\mathrm{mod}}$ is the modulation classification loss, $\mathcal{L}_{\mathrm{teacher}}$ is a Jensen–Shannon [31] alignment between student and teacher masks, $\mathcal{L}_{\mathrm{jammer}}$ and $\mathcal{L}_{\mathrm{link}}$ constrain the jammer-family and link-quality auxiliary objectives, $\mathcal{L}_{\mathrm{contrast}}$ is a supervised-contrastive loss [32] over exact-source positives that encourages condition invariance, $\mathcal{L}_{\mathrm{orth}}$ is a route-orthogonality regularizer, and $\alpha, \beta, \gamma, \eta, \theta$ are nonnegative weights. The exact-source contrastive loss is an auxiliary training objective, not a headline contribution.

TABLE II
INFORMATION CONTRACT ACROSS STAGES.

Quantity	Training	Test inference	Post-hoc analysis
Received mixture	Yes	Yes	Yes
Tracked target component	Teacher only	No	Simulator audit
Tracked jammer component	Teacher only	No	Overlap covariate
Clean counterfactual	No	No	Tier-2 probe only
True SIR	Simulator metadata	No	Envelope covariate

TABLE III
NOTATION AND MODEL IDENTIFIERS.

Identifier	Meaning
A0	direct spectral reference backbone
A5 / VIMD-Net	sealed full configuration
A7	residual-free control variant
S-tier / M-tier / L-tier	prospective spectral capacity tiers (five-seed subset)
sidecar	Tier-2 received-I/Q repair model
o	target-support jammer overlap, Eq. (9)
κ	jammer scaling, Eq. (1); λ, ρ: route gate/bypass
CSSL	adaptation-oriented reference [15], [33]
IQFormer-inspired	unified-retraining strong I/Q reference [7]
macro-F1	unweighted mean of class-wise F1

E. Information Contract and Complexity

Table II states which quantities are available at each stage, to remove any oracle-information ambiguity. VIMD-Net uses 39500 parameters and 41.8 million reported MACs. Table III fixes the model identifiers used throughout.

F. Implementation Details

Unless otherwise stated, all models are trained on the same source-disjoint cache with the same data budget, optimization budget, stopping rule, and validation-only model-selection criterion. Model-specific architectural and objective components follow the predefined configuration of each variant.

Input and front end. One example is a complex baseband window of 1024 samples (1024 μs at 1 MHz, 4 samples per symbol, root-raised-cosine roll-off 0.35), power-normalized by the frozen receiver automatic gain control so that every window carries the same average power. The spectral front end applies a two-sided STFT with $N_{\mathrm{FFT}} = 64$, hop 16, a periodic Hann window of length 64, and no centering or zero padding, giving a 64×61 time–frequency lattice at 15.6 kHz per bin. The student and the fixed teacher share this lattice exactly.

Architecture width. The sealed configuration uses 24 spectral channels, embedding dimension 48, environment dimension 32, and dropout 0.05, for 39500 parameters. The prospective capacity ladder widens only these three numbers: the M tier uses (40, 80, 48) and the L tier (64, 128, 64).

Optimization and model selection. All fits use AdamW with an initial learning rate of 3×10^{-4}, weight decay 10^{-2}, batch size 16, mixed precision, and gradient-norm clipping at 5.0, for a budget of 30 epochs. Auxiliary terms are staged: mask supervision starts at epoch 2 and the contrastive term at epoch 5, each ramped linearly over 3 epochs, so the complete objective is active from epoch 8. The learning rate is held at its initial value until the objective has been complete for 3 further epochs and then follows a cosine decay to a floor of 0.05 times the initial value at epoch 30. Checkpoints become selection-eligible at epoch 10; the reported checkpoint is the eligible epoch with the lowest validation modulation cross-entropy (view 1, no label smoothing, no auxiliary terms, strict-improvement tolerance 10^{-5}), with early stopping after 8 eligible epochs without improvement. This rule, the optimizer settings, and the schedule are shared across every model; no architecture-specific hyperparameter sweep was performed for any comparator, and validation data are never used for evaluation.

Loss weights. In (7), $\alpha = 0.50$, $\beta = 0.25$, $\gamma = 0.05$, $\eta = 0.10$, and $\theta = 0.01$, with label smoothing 0.05 and contrastive temperature 0.12. The route-orthogonality regularizer is part of the A5 whole-configuration bundle (Table V) but is not isolated in a dedicated component contrast.

Seeds. The sealed campaign uses 10 algorithm seeds (17, 29, 43, 71, 101, 131, 173, 211, 257, 307) shared across all models, with the model-initialization seed shared by reported seed so that paired comparisons are seed-aligned. The capacity ladder uses the first five of these seeds.

IV. Evidence Protocol

A. Frozen Source-Disjoint Campaign

The artifact-locked campaign contains 12 model variants, 10 algorithm seeds, 120 fits, 11 evaluation splits, and 100000 training source sequences. Target classes are BPSK, PI/2-BPSK, QPSK, 8PSK, 16QAM, 64QAM, 256QAM, GMSK, CPFSK, and 4FSK, covering PSK, QAM, continuous-phase, and FSK families. Structured jammer families span tone, multitone, chirp, sweep, partial-band, pulse, comb, and out-of-taxonomy (cochannel and OFDM-like) modulated sources. TDL-A/C/D profiles are seen during training, while TDL-B/E profiles support held-channel evaluation. The evaluation grid spans eight SNR levels from -10 to 18 dB. The in-distribution split uses SIR levels -10 to 10 dB in 5 dB steps, whereas the hard-interference split spans SIR $\in \{-15, -10, -5, 0\}$ dB; the SIR $= +5$ and $+10$ dB cells therefore belong to the in-distribution split and do not enter the hard-interference or envelope-fitting cells. The fit-domain division (in-distribution plus hard-interference cells) is fixed in the exploratory analysis protocol before any envelope is fitted. Jammer family, speed, channel, combined OOD, receiver stress, and clean-retention splits share source identities across model comparisons. The receiver-stress split (ADC quantization and per-emitter synchronization) is retained in the artifact package and evaluated for the Tier-2 sidecar, but it does not enter the sealed or envelope claims of this paper and is not used in any reported model-selection claim; its per-split aggregates are included in the reproduction package rather than in the manuscript.

TABLE IV
Comparator and Complexity Summary (Hard-Interference Setting).

Model	Domain	Params	MACs	Seeds
A0	spectral	8458	6.3 M	10
A5 / VIMD-Net	spectral allocation	39500	41.8 M	10
sidecar	spectral + received I/Q	46794	43.1 M	10
MCLDNN	I/Q	406070	398.2 M	10
IQFormer-inspired	multimodal I/Q	354984	355.6 M	10
CSSL	I/Q (adaptation)	8631948	155.3 M	10

B. Comparator Fairness

Table IV summarizes the comparators. All models are retrained under the same source-disjoint protocol; “IQFormer-inspired” and “MCLDNN” denote architecture-faithful unified-retraining references, not exact reproductions of every original training recipe. Comparator fairness is enforced at the budget rather than at the tuning level: every model sees the same training source count, the same epoch budget, the same early-stopping patience and checkpoint criterion, and the same validation protocol. CSSL is retained as an adaptation-oriented reference rather than a state-of-the-art anchor; under this unified structured-interference retraining protocol it underperforms the direct spectral A0 baseline despite using about 8.6 million parameters, and we do not present it as a tuned upper bound. Its 8.63 million parameters but only 155.3 million MACs reflect an aggressive encoder downsampling and a large projection head inherited from the official 1024-sample RadioML topology, so parameter count alone is a poor cost proxy for this comparator. Figures label this comparator “IQFormer-insp.” so that no figure has to be read against the text; “A5” and “VIMD-Net” denote the same sealed configuration, and “CSSL” abbreviates the contrastive self-supervised learning reference of [15], whose released code snapshot [33] we retrain under our protocol.

C. Original Sealed Confirmatory Family

The sealed family contains three hard-interference contrasts: the complete configuration versus A0, the margin teacher versus a proportional-power teacher, and the tri-route model versus a dual-route control. Table V records what differs between A0 and A5 so that the whole-configuration contrast is not over-interpreted. The A5–A0 contrast is interpreted as a whole-configuration effect, not as an isolated causal effect of the teacher or route count, because it also changes capacity (about $4.7\times$ parameters) and bundles the multi-task and contrastive terms with the allocation.

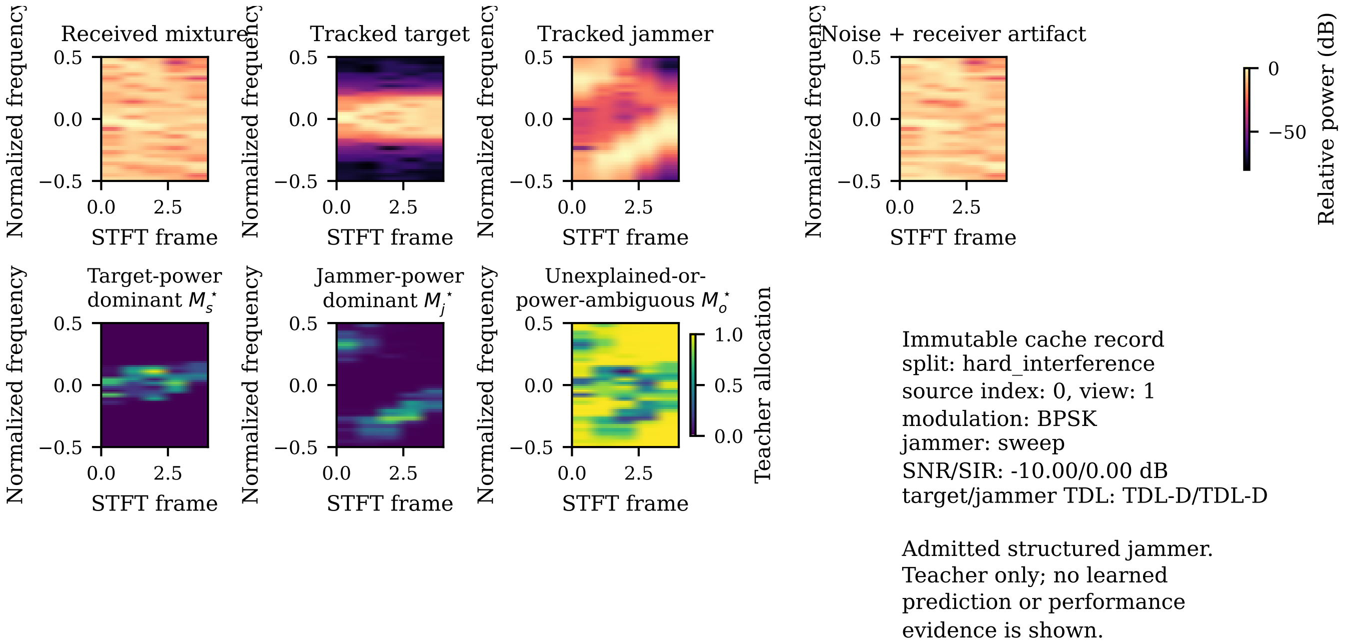

Fig. 1. A fixed-teacher proxy visualization from an immutable simulation record. The upper row shows the received mixture and tracked components; the lower row shows the three allocations on the same STFT lattice. The image explains the teacher definition and is not performance evidence.

TABLE V
WHAT DIFFERS BETWEEN A0 AND A5.

Property	A0	A5
Parameter count	8458	39500
Three-route allocation	No	Yes
Simulation-component teacher	No	Yes
Multi-task objectives	No	Yes
Exact-source contrastive	No	Yes
Residual bypass	No	Yes
Route-orthogonality regularizer	No	Yes

D. Exploratory and Tier-2 Evidence Classes

All SIR-cell, overlap, fusion, condition-transfer, probe, repair, and capacity analyses are exploratory. In particular, the severe SIR stratification is post hoc. Tier-2 experiments were designed after inspecting the frozen evidence and are outside the frozen confirmatory family; they cannot retroactively change a sealed conclusion. The clean-retention gate was preregistered separately and did not pass; clean behavior is therefore treated as a boundary to explain and repair, not as a preserved property of the sealed model.

E. Statistical Procedures

The sealed family uses a joint non-studentized max-absolute-deviation hierarchical paired bootstrap with 10000 draws, resampling class-stratified source clusters while keeping algorithm seeds as fixed blocks, so pairing is preserved at the source level and simultaneous intervals are computed from the joint max absolute deviation. The design resolution of 1.31 percentage points is the simultaneous minimum detectable effect at 80% power under this joint interval, and it is an exploratory calibration rather than a sealed significance threshold. Because the bootstrap keeps algorithm seeds as fixed blocks, the reported intervals are conditional on the realized seed set and do not propagate seed-to-seed training variance. As a complement, we computed a seed-randomization variant in which the ten realized seeds are resampled with replacement while the full evaluation split is retained: the A5–A0 contrast remains positive (95% percentile interval $[4.02, 5.19]$ percentage points) while the teacher-form and route-count contrasts span zero ($[-0.15, 0.42]$ and $[-0.15, 1.04]$ percentage points), so the qualitative conclusions do not depend on conditioning on the realized seed set. In the severe corner, all 10 seeds share the same sign; the sign-flip permutation probability 0.0018 is a permutation test over the source-paired samples (an exact two-sided seed-level sign test would give $2/1024 \approx 0.0020$). The SIR $= -15$ dB aggregate is computed from source-paired predictions pooled across the eight SNR conditions before macro-F1 aggregation; it is therefore not the arithmetic mean of the eight displayed cell-level differences in Fig. 2.

V. RESULTS

A. Positive Sealed Whole-Configuration Contrast and Bounded Component Effects

Table VI reports the complete sealed family. The complete configuration improves over its shared A0 backbone by 4.57 percentage points, with a simultaneous interval of $[3.65, 5.49]$ percentage points. This effect is about 3.5 times the design resolution of 1.31 percentage points. The teacher-form and route-count interventions do not clear zero individually; under the same joint interval their contributions are bounded within 1.36 percentage points. Thus the configuration-level effect is resolved but cannot be assigned to either isolated component at this resolution, and because the family rule required all three

TABLE VI
SEALED CONFIRMATORY HARD-INTERFERENCE FAMILY. DIFFERENCES ARE CANDIDATE MINUS REFERENCE IN MACRO-F1 PERCENTAGE POINTS.

Contrast	Diff.	Sim. low	Sim. high
Full configuration vs A0	4.57	3.65	5.49
Margin vs proportional teacher	0.16	−0.76	1.08
Tri-route vs dual-route	0.44	−0.47	1.36

contrasts, the family gate did not pass. At the same parameter tier, the residual-free A7 variant is nominally above A5 by 0.21 percentage points in the marginal hard-interference aggregate (Fig. 5); treating the ten realized seeds as a random effect gives a 95% percentile interval of $[-0.19, 0.62]$ percentage points, so the difference is not resolved at this design's power and we do not interpret it as a real effect; we therefore attribute the compact frontier position to the VIMD family at this budget, not uniquely to any single component.

B. Rank Exchange across the SNR–SIR Plane

Marginal hard-interference macro-F1 is 0.4406 for A5, 0.4599 for MCLDNN, and 0.4872 for IQFormer-inspired. Those averages favor the larger models, but Fig. 2 shows why no single ranking is adequate: rank changes with SNR and SIR, with A5 leading only in the severe, high-overlap corner.

At $\mathrm{SIR} = -15$ dB, in a post-hoc analysis, A5 gains 9.74 percentage points over MCLDNN ($[7.52, 12.14]$) and 10.46 percentage points over IQFormer-inspired ($[6.78, 14.29]$); these two headline intervals are marginal stratified-bootstrap intervals and are not multiplicity-adjusted. All 10 seeds have the same sign, and the sign-flip permutation probability is 0.0018. Of 64 inspected cells, 12 survive Holm correction; all are on the severe SIR row. These checks reinforce the corner result without converting the post-hoc analysis into confirmation, and the severe corner is an in-sample stratification of the hard-interference cells, not a held-out validation.

C. Empirical Overlap–SIR Operating Envelope

The rank exchange motivates an empirical envelope fit rather than another marginal average. For reference b, we fit

$$\Delta_b = \beta_{0,b} + \beta_{1,b} o + \beta_{2,b}\,\mathrm{SIR}, \tag{8}$$

where Δ_b is A5 minus the reference macro-F1 and o is the *target-support jammer overlap*, defined below. We use this name in place of the more common "spectral occupancy" because o is not an occupied-bin or occupied-bandwidth fraction: it measures where the jammer energy lands relative to the target, not how much of the plane the jammer fills.

The covariate is defined from the simulator-tracked target and jammer components rather than from the mixture. For each window we compute the STFT power of the tracked target and of the tracked jammer separately, order the time–frequency cells by descending target power, and take the target's 99%-energy support $\mathcal{S}$ as the smallest set of cells whose cumulative target power reaches 99% of the total. The overlap is then the fraction of the jammer's total STFT power that falls inside $\mathcal{S}$:

$$o = \frac{\sum_{(f,t)\in\mathcal{S}} P_j(f,t)}{\sum_{(f,t)} P_j(f,t)}. \tag{9}$$

Thus $o = 0.8$ means that 80% of the jammer's energy lies within the target's dominant support, not that the jammer fills 80% of the plane. The envelope covariate is the mean of this per-window fraction over the cells of an evaluation split.

Equation (9) is evaluated on a fixed analysis lattice, a Hann-windowed 128-point STFT with hop 32, that is deliberately independent of the 64-point VIMD front-end lattice of Section III-F. The two serve different purposes: the front-end lattice is what the model sees, whereas the analysis lattice is a measurement instrument for the physical audit and is given finer frequency resolution for that purpose. The overlap value is computed once per window when the cache is built, before any model is trained; it is stored in the sealed cache, checksummed in the cache manifest, and read unchanged by every model, split, and comparison in this paper. It is never an input feature of VIMD-Net or of any comparator, and it is not tuned against any model's predictions.

Only the 32 in-distribution and hard-interference cells enter the fit; the remaining 81 cells (unseen jammer, unseen speed, held-out channel, combined OOD, and clean retention) are held out. Jammer-free clean-retention windows carry no jammer power and the cache's invalid-SIR sentinel, normalized to 0 dB; the clean-retention cell therefore enters Eq. (8) with $(o, \mathrm{SIR}) = (0, 0)$. Excluding this single cell leaves the held-regime skill essentially unchanged (RMSE 6.23 percentage points, $R^2_{\mathrm{skill}} = 0.45$ for IQFormer-inspired; 8.10 percentage points, $R^2_{\mathrm{skill}} = 0.29$ for MCLDNN), so the clean cell does not drive the reported result. Overlap and SIR are nearly orthogonal across the fitting cells (Pearson $r = -0.003$; -0.03 over all 113 cells), so strong linear collinearity between the two fitted covariates is not evident. We nevertheless interpret Eq. (8) as descriptive rather than causal, because the overlap is simulator-derived and the study is observational with respect to this covariate; low pooled correlation bounds collinearity, not confounding.

Against IQFormer-inspired, the fitted overlap coefficient is 11.39 percentage points per unit overlap and the SIR coefficient is -0.666 percentage points per dB. Magnitude skill is scored against an explicitly defined trivial baseline: the constant predictor outputs the mean gain over the 32 fitting cells (-8.27 percentage points for IQFormer-inspired, -4.63 percentage points for MCLDNN) and is evaluated unchanged on the 81 held-regime cells, with $R^2_{\mathrm{skill}} = 1 - \mathrm{MSE}_{\mathrm{envelope}}/\mathrm{MSE}_{\mathrm{const}}$. On that basis the envelope attains RMSE 6.20 percentage points against 8.40 percentage points for the constant predictor, for $R^2_{\mathrm{skill}} = 0.46$ (Fig. 4); the MCLDNN envelope gives 8.05 percentage points against 9.63 percentage points ($R^2_{\mathrm{skill}} = 0.30$). We also fitted the three-variable variant $\Delta_b = \beta_{0,b} + \beta_{1,b} o + \beta_{2,b}\mathrm{SIR} + \beta_{3,b}\mathrm{SNR}$ that adds the cell-level mean SNR. It lowers the held-regime RMSE only modestly (RMSE 6.12 percentage points, $R^2_{\mathrm{skill}} = 0.47$ for IQFormer-inspired; RMSE 8.00 percentage points, $R^2_{\mathrm{skill}} = 0.31$ for MCLDNN), so we retain the two-variable

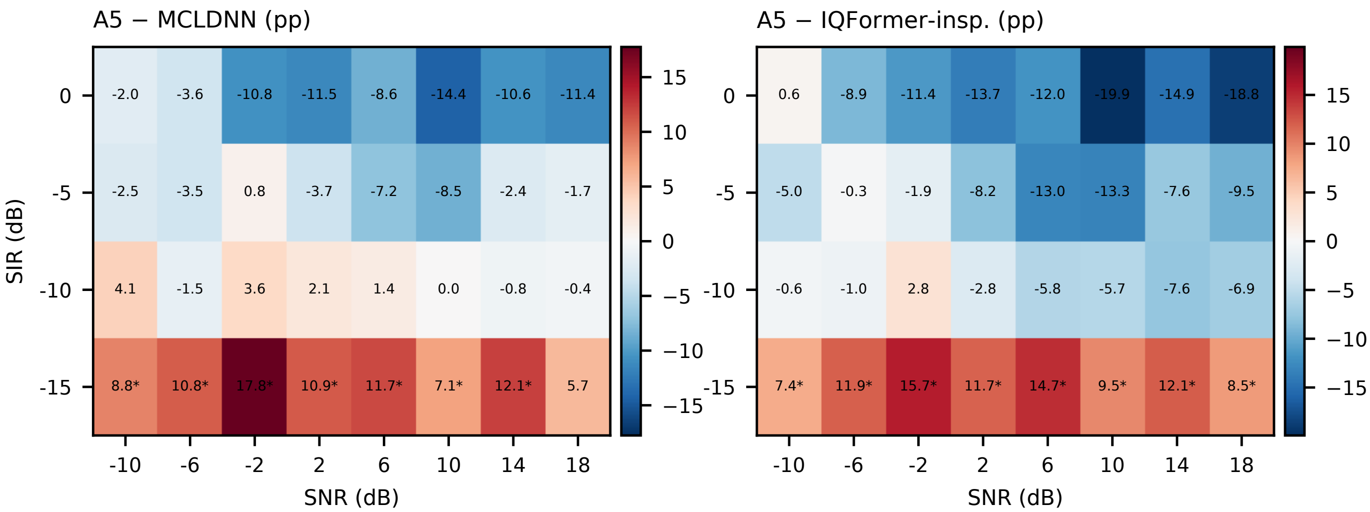

Fig. 2. Post-hoc operating map over all hard-interference SNR–SIR cells. Each entry is A5 minus the named strong baseline in macro-F1 percentage points, rounded to one decimal; a star marks a positive unadjusted paired lower interval, while Holm-surviving cells are reported separately in the text. Displaying the full map prevents the severe corner from being read as a uniform ranking.

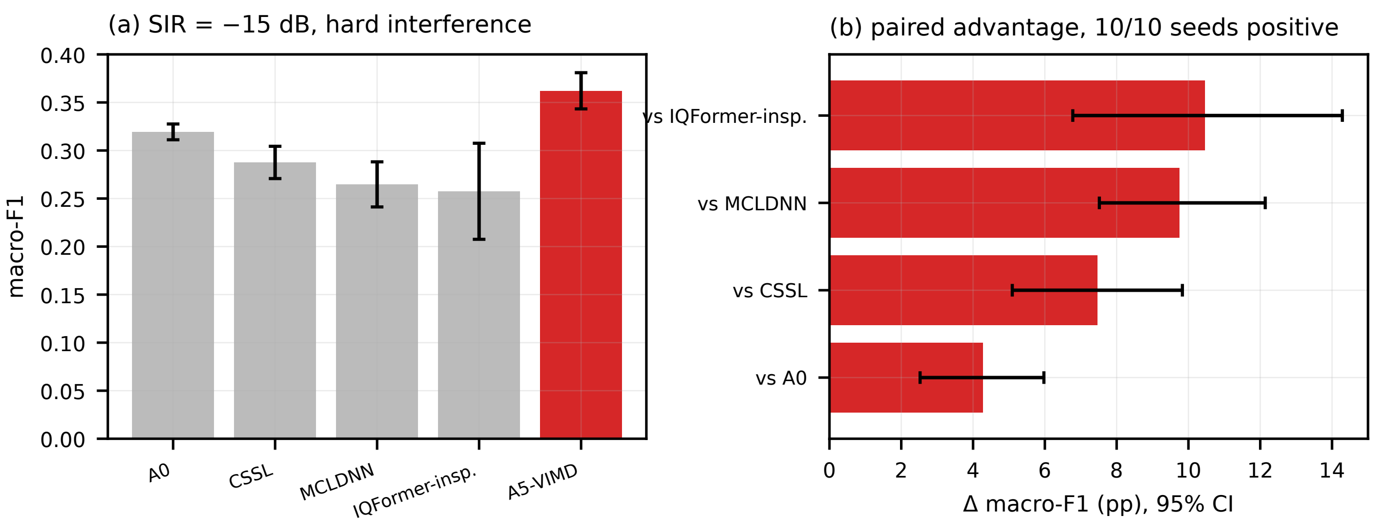

Fig. 3. Severe-interference corner. Left: absolute macro-F1 at SIR $= -15$ dB (mean over the ten seeds; error bars are the seed standard deviation). Right: source-paired A5 advantages with stratified bootstrap intervals. The comparison is exploratory and post hoc.

model as primary and report the three-variable variant as a sensitivity check. Two limits of this score are stated rather than left to the reader. First, an oracle constant equal to the mean of the held-regime cells themselves would attain 6.03 and 6.77 percentage points, corresponding to $R^2_{\text{skill}} = -0.05$ and -0.41 against that same reference: the envelope is worse than a hindsight-optimal constant within the held-regime domain, so the envelope's advantage lies in transporting the level of the gain across regimes rather than in resolving variation within them. Second, sign agreement is 0.975 (0.951 for MCLDNN) while 0.963 (0.901) of held-regime gains are negative, so a majority-sign predictor is nearly as accurate; sign agreement is reported as a secondary diagnostic only.

The skill is therefore not uniform: it is strongest in the unseen-speed and held-out-channel regimes and weak or negative in the unseen-jammer and combined-OOD regimes, so the pooled R^2_{skill} is a regime-weighted average rather than a uniformly strong transport property.

Solving Eq. (8) for zero gain yields the break-even overlaps in Table VIII. Each entry answers a single question: at a given SIR, what fraction of the jammer's energy must land inside the target's dominant support before the predicted gain of A5 over the reference crosses zero? The break-even value is therefore a simulator-domain jammer-to-target overlap threshold, not an occupied-bandwidth threshold. Because the overlap is a simulator-tracked quantity, these are reference values conditional on an overlap estimate, not deployment-ready control thresholds, and they should be read only within the fitted SIR range; extrapolation beyond that range is not evaluated. The SIR $= -10$ dB entry against IQFormer-inspired already exceeds the largest per-cell overlap observed in this campaign (0.84), so it marks a break-even point the campaign does not actually reach rather than one it verifies. The required overlap rises as interference becomes less severe.

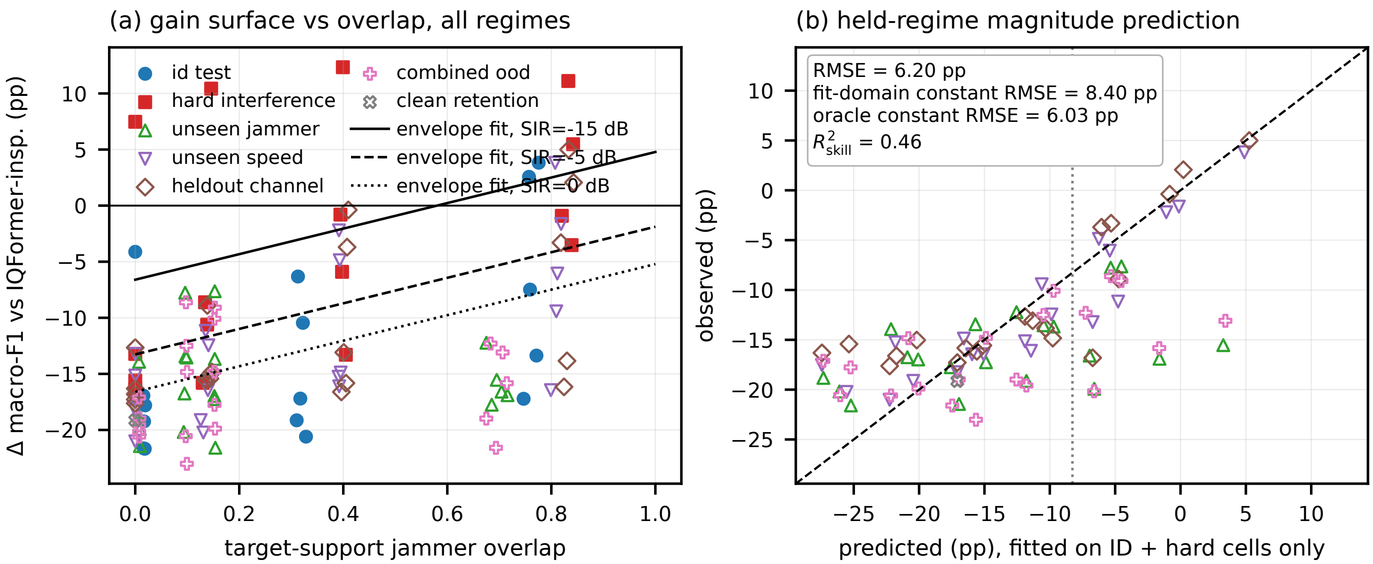

Fig. 4. Exploratory overlap–SIR operating envelope. The abscissa of panel (a) is the target-support jammer overlap of Eq. (9), not an occupied-bandwidth fraction. Filled cells are used to fit Eq. (8); open cells denote regime cells excluded from fitting. Panel (b) compares predicted and observed model gains on the held-regime cells. Magnitude prediction is evaluated by RMSE relative to a constant predictor estimated from the envelope-fitting cells (whose value is the mean gain there, -8.27 percentage points): 6.20 percentage points versus 8.40 percentage points for the constant ($R^2_{\text{skill}} = 0.46$); an oracle constant equal to the mean of the held-regime cells themselves would attain 6.03 percentage points. Sign agreement is reported only as a secondary diagnostic, because the held-regime sign distribution is strongly imbalanced.

TABLE VII
HELD-REGIME ENVELOPE SKILL BY REGIME. GAINS AND RMSE IN PERCENTAGE POINTS; THE CONSTANT BASELINE IS THE MEAN GAIN OVER THE 32 FITTING CELLS. THE CLEAN-RETENTION ROW IS A SINGLE CELL AND IS REPORTED FOR COMPLETENESS ONLY.

Reference	Held regime	Cells	Obs. gain	Env. RMSE	Const. RMSE	R^2_{skill}
IQFormer-insp.	unseen speed	20	−12.4	3.94	7.87	0.75
IQFormer-insp.	held channel	20	−11.3	4.86	7.59	0.59
IQFormer-insp.	combined OOD	20	−16.5	7.50	9.23	0.34
IQFormer-insp.	unseen jammer	20	−16.1	7.74	8.67	0.20
IQFormer-insp.	clean retention	1	−19.1	2.06	10.87	0.96
IQFormer-insp.	pooled	81	−14.1	6.20	8.40	0.46
MCLDNN	unseen speed	20	−8.9	4.66	8.88	0.73
MCLDNN	held channel	20	−8.0	5.41	9.06	0.64
MCLDNN	combined OOD	20	−14.6	10.23	10.28	0.01
MCLDNN	unseen jammer	20	−14.4	10.34	10.24	−0.02
MCLDNN	clean retention	1	−14.1	0.46	9.46	1.00
MCLDNN	pooled	81	−11.5	8.05	9.63	0.30

TABLE VIII
EXPLORATORY BREAK-EVEN TARGET-SUPPORT JAMMER OVERLAP.

SIR (dB)	vs IQFormer-inspired	vs MCLDNN
−15	0.58	0.48
−10	0.87	0.66

The preregistered mechanism expectation was non-increasing gain with overlap. The sealed directional test rejected that sign. Exploratory quintiles instead show monotonically increasing A5–CSSL gain, from a negative low-overlap region to a strongly positive high-overlap region. We return to the interpretation of this reversal in Section VI-C.

D. Complexity, Complementarity, and Selective Prediction

A5 remains useful outside its winning region because its errors differ from those of the I/Q models. Probability fusion with IQFormer-inspired improves over the stronger constituent by 4.71 percentage points, whereas a same-architecture seed-ensemble control improves by only 1.49 percentage points; the corresponding MCLDNN fusion and control gains are 5.70 and 1.69 percentage points. This is supporting evidence of complementary decisions, not evidence that the compact model alone wins the marginal benchmark.

A5 uses 39500 parameters, 41.8 million MACs, and a median latency of 3.172 ms, versus 354984 parameters, 355.6 million MACs, and 4.180 ms for IQFormer-inspired, and 406070 parameters, 398.2 million MACs, and 3.789 ms for MCLDNN. The compact advantage is thus strongest in parameter count and MACs; the median-latency advantage over MCLDNN is modest, so we frame compactness as a parameter and MACs property rather than an end-to-end inference-cost claim. At 30% retained coverage after validation-only confidence calibration, A5 achieves 94.4% selective accuracy, close to the IQFormer-inspired 94.9% (Fig. 6). Selective prediction is supporting rather than central evidence.

E. Clean-Boundary Diagnosis

The clean-retention split contains all ten modulations, but the two training views expose jammer-free windows for only 4 classes (PI/2-BPSK, 8PSK, 256QAM, and CPFSK); the remaining 6 classes (BPSK, QPSK, 16QAM, 64QAM, GMSK, and 4FSK) are a condition-transfer test. The four covered classes are marked without a star in Fig. 7, where a star denotes a class whose jammer-free condition was not seen during training. Among uncovered constellation-order cells, transfer failure occurs in 100% of compact-spectral cells versus 11% of I/Q high-capacity cells. The clean-retention gate did not pass. Figure 7 localizes the failure rather than hiding it in an aggregate.

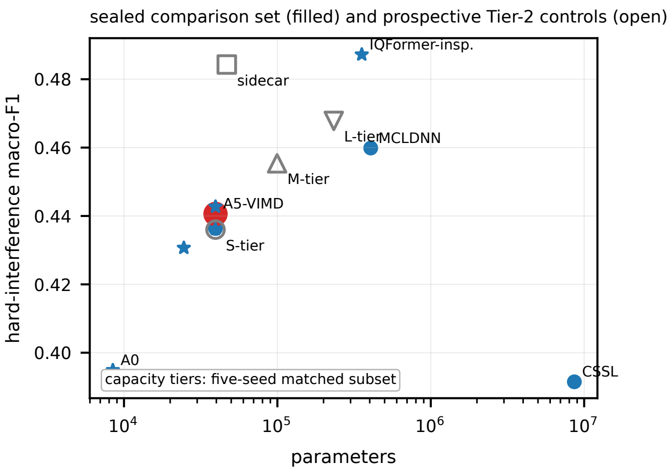

Fig. 5. Hard-interference macro-F1 versus parameter count. Filled markers denote the sealed comparison set and the Tier-2 sidecar with their ten-seed means; open markers denote the prospective capacity tiers, which are evaluated on the five-seed matched subset and are therefore not numerically mixed with the ten-seed points (Table X). The Pareto reading is descriptive across evidence classes and does not merge the exploratory models into the sealed family. The frontier is also regime-specific.

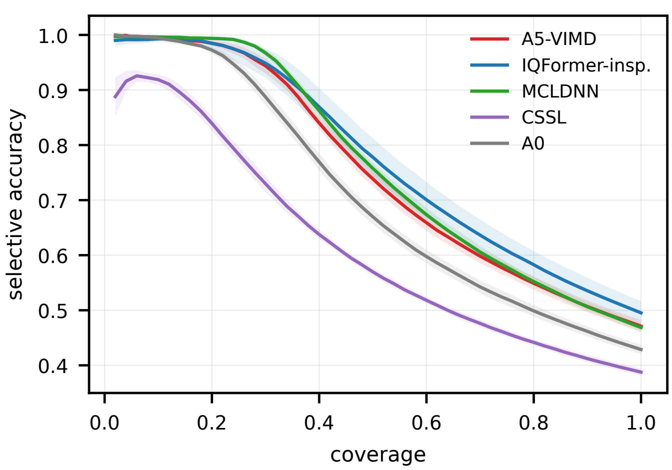

Fig. 6. Selective accuracy versus coverage on hard interference. Thresholds are calibrated without using test labels. This is supporting evidence.

We then followed a prospective decision chain (Table IX). Frozen-checkpoint linear and MLP probes used a correctly reconstructed jammer-free counterfactual and source-disjoint clean cross-validation. Probe performance is the unweighted mean F1 over three constellation-order-sensitive classes (QPSK, 16QAM, and 64QAM), ranging over $[0, 1]$, and is close to 0.1 for chance on this ten-class problem. The decision rule, preregistered in the Tier-2 protocol before any probe was fitted, declares a representation to be *information-present, head-limited* when this mean exceeds 0.25 and *representation-limited* otherwise; 0.25 was set as roughly the midpoint between chance and the level a usable constellation-order code would have to reach, and the same threshold applies to the linear and MLP probes. For A5, 5 of 6 design–seed cells lie below it, and one cell is borderline (0.256). The six design–seed cells comprise three algorithm seeds $\times$ two probe designs (the correctly reconstructed jammer-free counterfactual and source-disjoint clean cross-validation), evaluated with the MLP probe. The preregistered consistency rule declares the representation information-present only if *all* cells exceed the threshold, so with five of six below we record the representation-limited side. The determination is robust to raising the threshold to 0.30 (all six cells then lie below it) but flips if it is lowered to 0.20 (all six then exceed it), so we state this sensitivity explicitly. The strong I/Q baselines exceed the threshold in 18 of 18 cells. The evidence therefore leans toward missing constellation information in the compact spectral representation rather than a classification head that fails to use available information.

Adding clean windows for every class tests the simpler coverage explanation. It changes clean macro-F1 by only 0.15 percentage points ($[-0.31, 0.63]$) and hard macro-F1 by -0.30 percentage points ($[-0.91, 0.27]$). Coverage alone is therefore insufficient. A preregistered negative control gates the modulation route using predicted interference presence. Relative to A5, it changes clean macro-F1 by -0.30 percentage points ($[-0.66, 0.08]$) and hard macro-F1 by -0.18 percentage points ($[-0.72, 0.35]$). The learned gate averages 99.994% on clean windows and 99.991% under hard interference, so it saturates near the modulated route in both conditions rather than implementing the intended clean identity fallback. This arm is a negative control, not stand-alone evidence that a functional mask fallback was learned.

F. Prospective I/Q Repair and Capacity Controls

The next intervention adds a received-I/Q sidecar while retaining the spectral routes and the same source mixture. It increases the *total* model size from 39500 to 46794 parameters (+18%) and the MAC count to 43.1 million (about +3%); the sidecar is a small addition to the model, not a 46794-parameter module bolted onto a 39500-parameter one. It raises clean macro-F1 by 11.77 percentage points ($[11.06, 12.49]$) and hard macro-F1 by 4.37 percentage points ($[3.44, 5.31]$), i.e. clean macro-F1 rises from 0.5005 to 0.6182 and hard-interference macro-F1 from 0.4406 to 0.4843. Its overall hard score is statistically near IQFormer-inspired ($\Delta = -0.29$ percentage points, $[-1.95, 1.46]$), while at severe SIR (-15 dB) it remains ahead of IQFormer-inspired by 12.09 percentage points ($[8.32, 15.88]$). The sidecar was designed prospectively after the clean-boundary diagnosis; it is interpreted as an intervention on the hypothesized information bottleneck, not as retroactive evidence for the original A5 architecture and not as a new multimodal fusion architecture. This repairs the clean boundary without erasing the structured-interference region.

The sidecar is a lightweight raw-I/Q temporal branch attached to the sealed A5 configuration: a two-channel 1-D depthwise-separable convolutional network ($2 \to 12 \to 12$ depthwise$\to 24 \to 24$ depthwise channels) with group normalization and SiLU activations, whose per-window temporal mean, standard deviation, and maximum are pooled and projected to a 24-dimensional embedding; this embedding is fused with the spectral embedding at feature level and the fused readout is added to the logits as a residual. The branch adds 7,294 parameters (39,500$\to$46,794, +18%) but only about

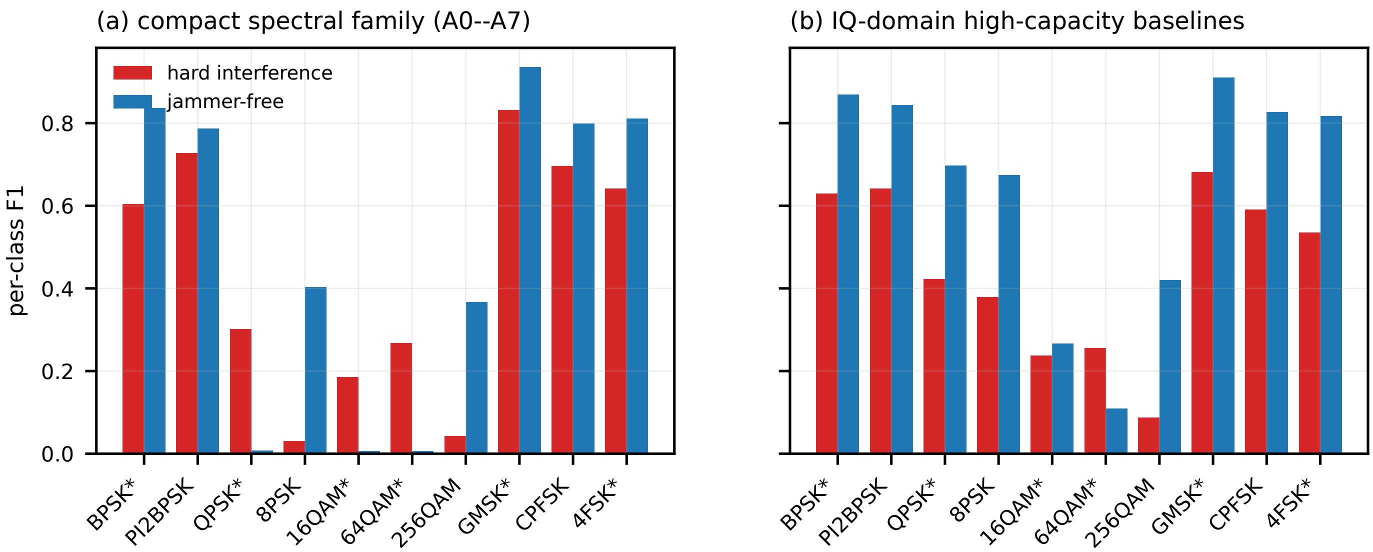

Fig. 7. Condition-transfer taxonomy. A star marks a class whose jammer-free condition was not seen during training (jammed-only coverage); the four covered classes (PI/2-BPSK, 8PSK, 256QAM, CPFSK) are unstarred. Panel (a) pools the compact spectral family (A0–A7) as per-class mean F1, which is the family-level claim made in the text. Training coverage and discriminability type separate the compact spectral failure from classes whose envelope cues remain distinctive.

1.3 million MACs (41.8M→43.1M, +3%), because it is a single low-temporal-resolution pass with depthwise kernels and aggressive time pooling, while the parameter growth is dominated by the projection and fusion layers. It consumes only the received mixture and requires no component book-keeping at inference.

Finally, we widen the spectral model without changing its representation domain. The S-tier, M-tier, and L-tier contain 39500, 99596, and 233244 parameters. Relative to the S-tier, the M-tier improves hard macro-F1 by 1.94 percentage points and the L-tier by 3.18 percentage points; the L-tier improves over the M-tier by 1.24 percentage points. Capacity therefore improves average fitting. The capacity ladder uses a five-seed matched subset for both candidates and references, so its comparator means differ from the ten-seed main results by up to 1.36 pp: the five-seed S-tier minus IQFormer-inspired is -6.02 percentage points versus -4.66 percentage points for the ten-seed A5–IQFormer-inspired comparison, a gap of the same order as the design resolution and attributable to the seed-subset difference rather than to any model change. Table X uses the matched subset throughout; for reference, the five-seed absolute macro-F1 values are S-tier 0.436, M-tier 0.455, L-tier 0.468, IQFormer-inspired 0.496, and MCLDNN 0.454. The L-tier remains below IQFormer-inspired on overall hard interference, with a difference of -2.84 percentage points ($[-4.30, -1.31]$), so the remaining gap is not attributable to capacity. At SIR $= -15$ dB, the same L-tier model is ahead of IQFormer-inspired by 14.07 percentage points ($[9.55, 18.51]$); this five-seed comparison is not directly comparable to the ten-seed A5 severe-corner gain of 10.46 percentage points. Increasing spectral capacity therefore shifts performance but does not remove the operating boundary, supporting a representation interpretation rather than a pure parameter-count explanation. The ladder varies width only: S is the sealed A5 configuration itself (24 spectral channels, embedding 48, environment 32), and M and L widen exactly those three numbers with no change to the routes, teacher, objective, or training recipe. The S row therefore reuses the sealed A5 checkpoints; its hard-interference score differs from the main table only because the ladder is evaluated on the five-seed matched subset, not because the model differs.

VI. Discussion

A. What the Operating Envelope Establishes

The evidence supports a practical model-selection statement within the simulator domain. Within the present simulator campaign, the compact spectral configuration becomes relatively most competitive when interference is severe and jammer energy strongly overlaps the target-dominant time–frequency support. This empirical pattern should not be interpreted as confirmation of a specific routing or allocation mechanism; no sealed contrast isolates the teacher or route component (Section V-A). A larger I/Q model is preferable in weak, clean, or low-overlap conditions unless the compact system includes an I/Q side channel. The two models are also complementary enough that late probability fusion can outperform either alone. Overlap and SIR show little linear correlation in the fitted cell set, indicating that the envelope is not simply reproducing one covariate through the other; this does not establish causal independence.

B. What It Does Not Establish

Eq. (8) is empirical and descriptive, not a causal structural equation. The target-support overlap is simulator-derived because it requires separately tracked target and jammer components, and it is not directly observable at an uncooperative receiver; the break-even overlaps of Table VIII are therefore simulator-domain reference values conditional on an overlap estimate. The severe SIR corner is an in-sample stratification,

TABLE IX
PROSPECTIVE TIER-2 DIAGNOSIS, REPAIR, AND NEGATIVE CONTROL. THESE RESULTS ARE EXPLORATORY AND OUTSIDE THE FROZEN CONFIRMATORY FAMILY.

Intervention	Question	Params	MACs	Clean diff. [95% CI]	Hard diff. [95% CI]	Decision
Per-class clean coverage	Is exposure alone sufficient?	39.5k	41.8M	0.15 [−0.31,0.63]	−0.30 [−0.91,0.27]	No
Presence-gated route vs A5	Is mask application the primary clean cause?	39.5k	41.8M	−0.30 [−0.66,0.08]	−0.18 [−0.72,0.35]	No; $g \approx 1$ in both conditions
Received-I/Q sidecar vs A5	Is the representation gap repairable?	46.8k	43.1M	11.77 [11.06,12.49]	4.37 [3.44,5.31]	Yes; both intervals positive

TABLE X
EXPLORATORY CAPACITY LADDER ON HARD INTERFERENCE. ALL CANDIDATE AND COMPARATOR VALUES ARE RECOMPUTED ON THE SAME FIVE-SEED MATCHED SUBSET AND SHOULD NOT BE NUMERICALLY MIXED WITH THE TEN-SEED MAIN-RESULT MEANS.

Contrast	Diff.	Low	High
M-tier minus S-tier	1.94	1.07	2.80
L-tier minus S-tier	3.18	2.44	3.96
L-tier minus M-tier	1.24	0.31	2.37
L-tier minus IQFormer-inspired	−2.84	−4.30	−1.31

and the held-out evaluation covers regimes within the same frozen simulator campaign, not real-world out-of-distribution conditions.

C. Meaning of the Failed Directional Hypothesis

The preregistered directional mechanism hypothesis was not supported. The data reject the expectation that the compact model's relative gain should decrease with increasing overlap. The surviving result is therefore empirical rather than mechanistic: the target-support jammer overlap organizes the observed rank exchange, but the observed direction should not be interpreted as confirmation of the originally hypothesized masking mechanism. In this paper, "physics-informed" refers to the construction of the training supervision, not to a confirmed monotonic physical law.

D. Representation Repair and Practical Model Selection

The sequence of interventions matters. The frozen probe narrows the cause; the coverage arm rejects a cheap explanation; the I/Q sidecar repairs the predicted missing information; and the capacity ladder shows that width alone does not eliminate the boundary. The Tier-2 evidence is more consistent with a representation-information gap than with insufficient clean exposure alone. The sidecar is a prospective intervention on that hypothesis, not the new main classifier and not a multimodal-fusion novelty.

E. External Validity and Deployment Proxies

A future prospective campaign should randomize overlap and SIR more independently and estimate a receiver-observable proxy for jammer-to-target time–frequency overlap, potentially using occupied-bandwidth, spectral-kurtosis, interference-localization, or learned overlap estimators, then test whether the resulting overlap–SIR envelope retains cross-regime transport skill without simulator component bookkeeping. The present campaign establishes an internally controlled operating boundary; validation with receiver-observable overlap estimates and over-the-air or public recordings [34], [35] is a separate external-validity step. Public benchmarks would not, however, validate the envelope itself, since they generally do not expose the separately tracked target and jammer components required to compute the present target-support overlap covariate.

F. Vehicular Relevance of the Operating Boundary

The operating boundary maps onto concrete ITS conditions without new experiments. At $f_c = 5.9$ GHz (Section III-A), the break-even overlap of Table VIII (0.58 against IQFormer-inspired at SIR $= -15$ dB) describes a jammer whose energy is largely co-located with the target's dominant time–frequency support. That configuration arises in several ITS settings: adjacent-channel and same-band leakage in dense ITS-G5 roadside deployments, co-channel emitters in tunnel or canyon geometries where multipath spreads both signals over the same support, and periodic safety messages from other vehicles that occupy the target's band during high-traffic epochs. In these conditions the compact spectral front end is relatively most competitive. Conversely, in low-traffic or clean scenarios the overlap is near zero and a high-capacity I/Q path is preferable unless an I/Q side channel is present. The envelope therefore parameterizes, in simulator terms, when a vehicular receiver should switch between a cheap spectral path and an expensive I/Q path; the practical reading is conditional on an observable overlap estimator (Section VI-E), and the boundary itself is a simulation-domain design rule rather than a deployment-ready controller.

VII. LIMITATIONS AND REPRODUCIBILITY

The evidence has the following explicit limits. First, all results are from source-disjoint TR 38.901 TDL-profile simulations, not a complete vehicular geometry or a MAC-/network-layer V2X evaluation. Second, the teacher uses simulation-component bookkeeping and cannot supervise unreferenced recordings. Third, the target-support overlap covariate is simulator-derived because it requires separately tracked target and jammer components, and is not directly available at a receiver. Fourth, the original confirmatory family gate did not pass; only the complete A5–A0 contrast is a positive sealed result. Fifth, the clean-retention gate did not pass. Sixth, the severe-interference, operating-envelope, fusion, condition-transfer, probe, sidecar, and capacity analyses are exploratory or post-hoc, with post-hoc status stated where applicable. Seventh, public or over-the-air external validation is not yet established. Eighth, the IQFormer-inspired and MCLDNN comparators are unified-retraining references, not exact reproductions of every original training recipe.

Reproducibility is enforced at the artifact boundary. Each manuscript macro stores the source path, row selector, evidence class, precision, and source SHA-256; the resolved macro appendix in this submission exposes the numerical values and evidence class. The full provenance manifest, resolved macros, figure-source CSV files, per-seed aggregates, and run configurations are released as a reproduction package upon acceptance. Figures are rendered from CSV files and carry a separate provenance manifest. The frozen run stores cache identity, configurations, checkpoints, source-aligned predictions, seed aggregates, paired statistics, environment metadata, and execution records. These controls prevent a favorable Tier-2 result from overwriting an unfavorable sealed gate.

VIII. Conclusion

Compact spectral and high-capacity I/Q representations exchange rank across the structured-interference plane. A low-dimensional overlap–SIR envelope transports the gain level across regime cells excluded from fitting with measurable skill relative to a fit-domain constant baseline, while a post-hoc severe-interference analysis identifies a reproducible compact-model advantage in one corner of the plane. The original sealed family resolves a positive whole-configuration A5–A0 contrast but not the isolated teacher or route-count effects. A prospective probe–coverage–sidecar chain further indicates that the clean-condition failure is repairable by restoring a small amount of received-I/Q information, while spectral capacity alone does not remove the boundary; the intervention chain localizes a repairable representation bottleneck within the tested architecture family rather than establishing a causal diagnosis. The result is therefore a simulation-domain operating envelope and representation-repair study, not a claim of uniform superiority or a deployment-ready selector.

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